Homework 1

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1 Task 1

1.1 AR(1) without Stochastic Component

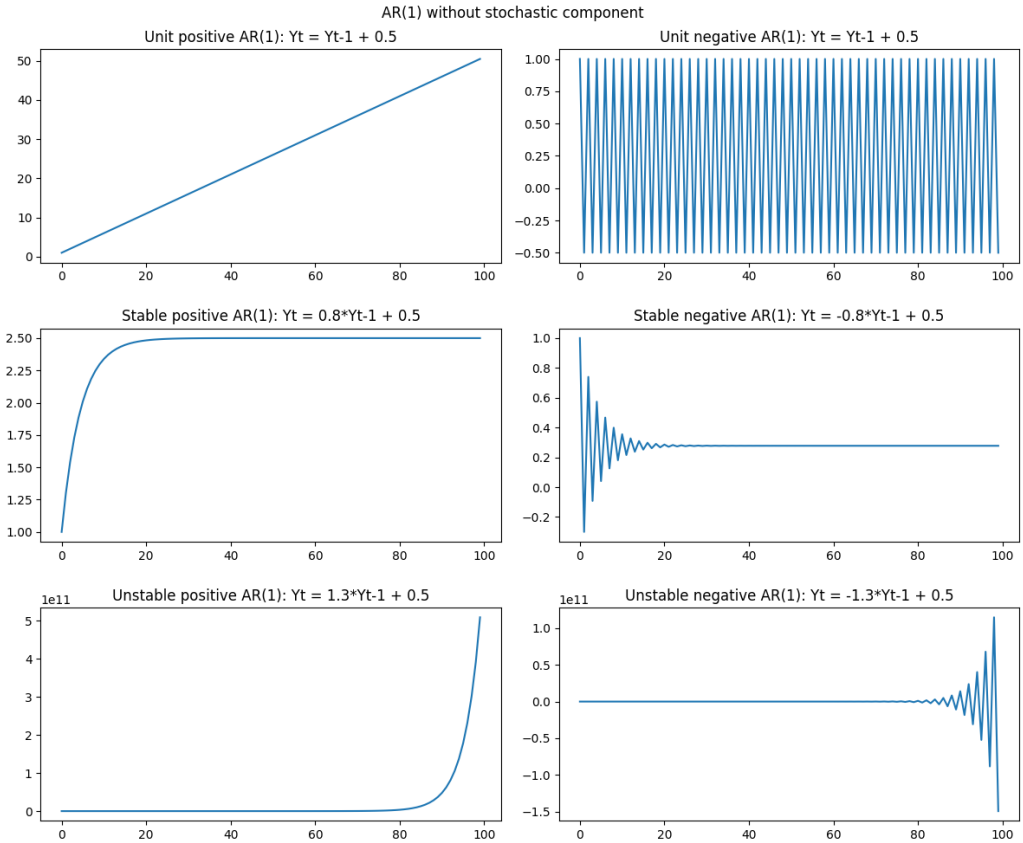

Figure 1: AR(1) process without stochastic component

1.2 AR(1) with Stochastic Component

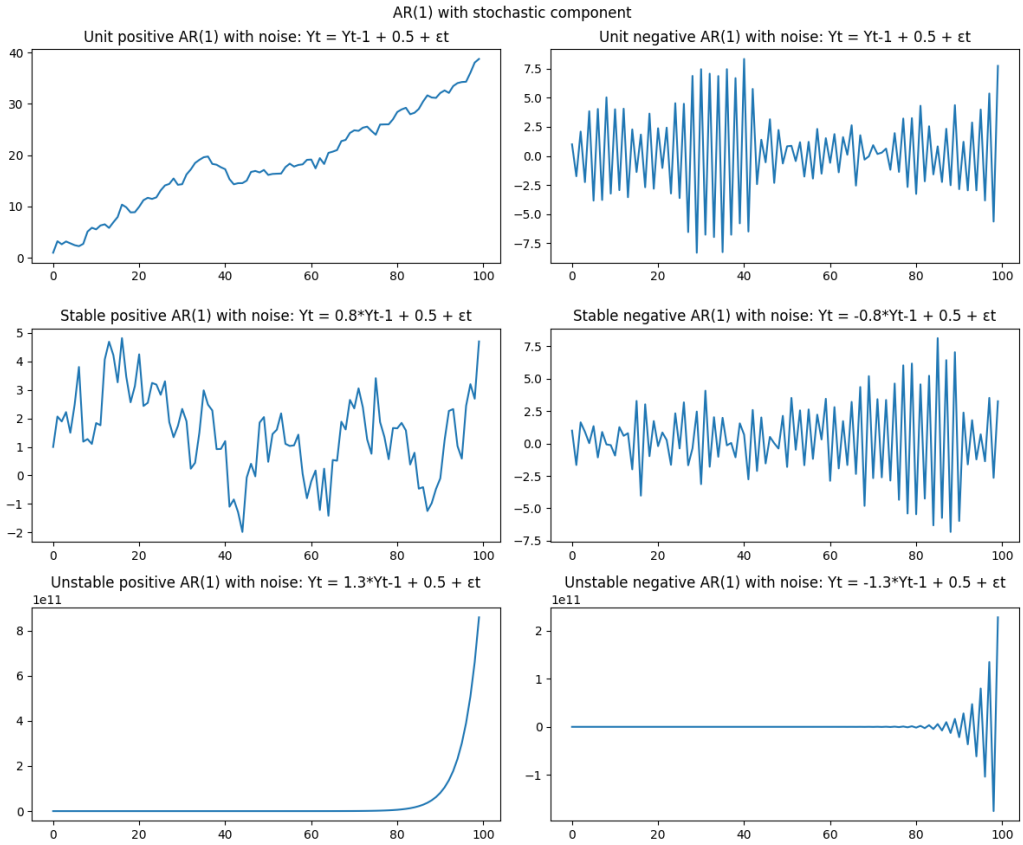

Figure 2: AR(1) process with stochastic component

1.3 AR(2) without Stochastic Component

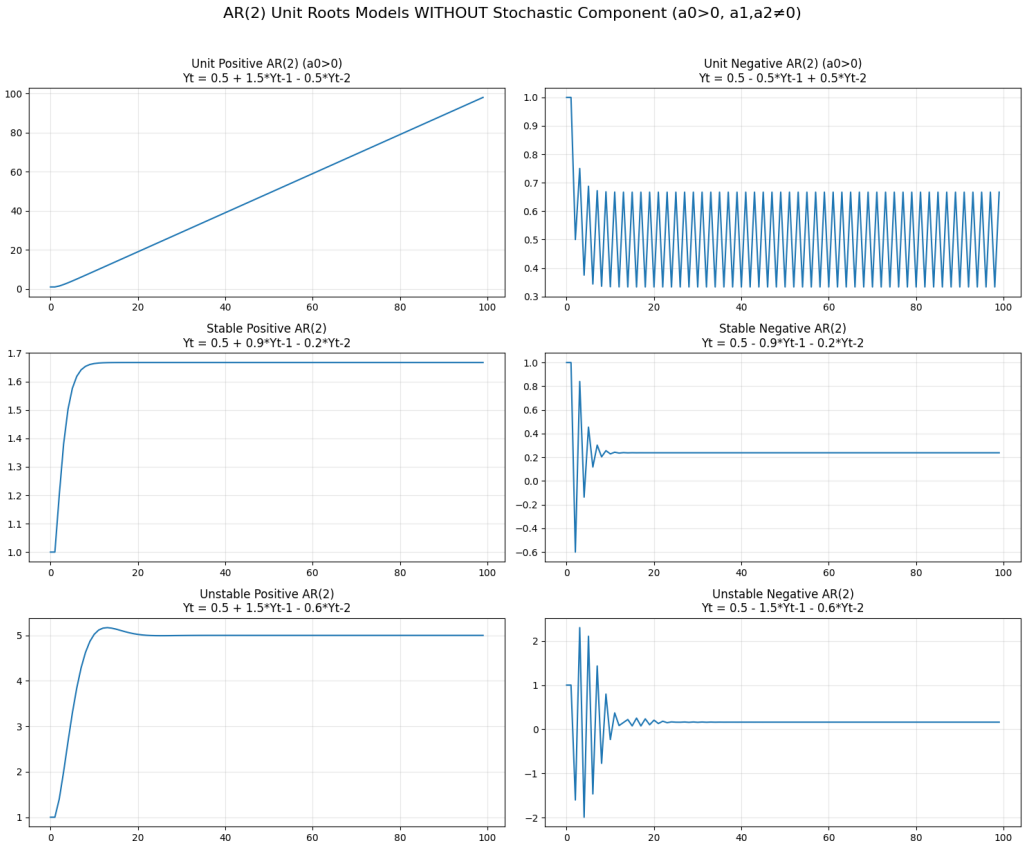

Figure 3: AR(2) process without stochastic component

1.4 AR(2) with Stochastic Component

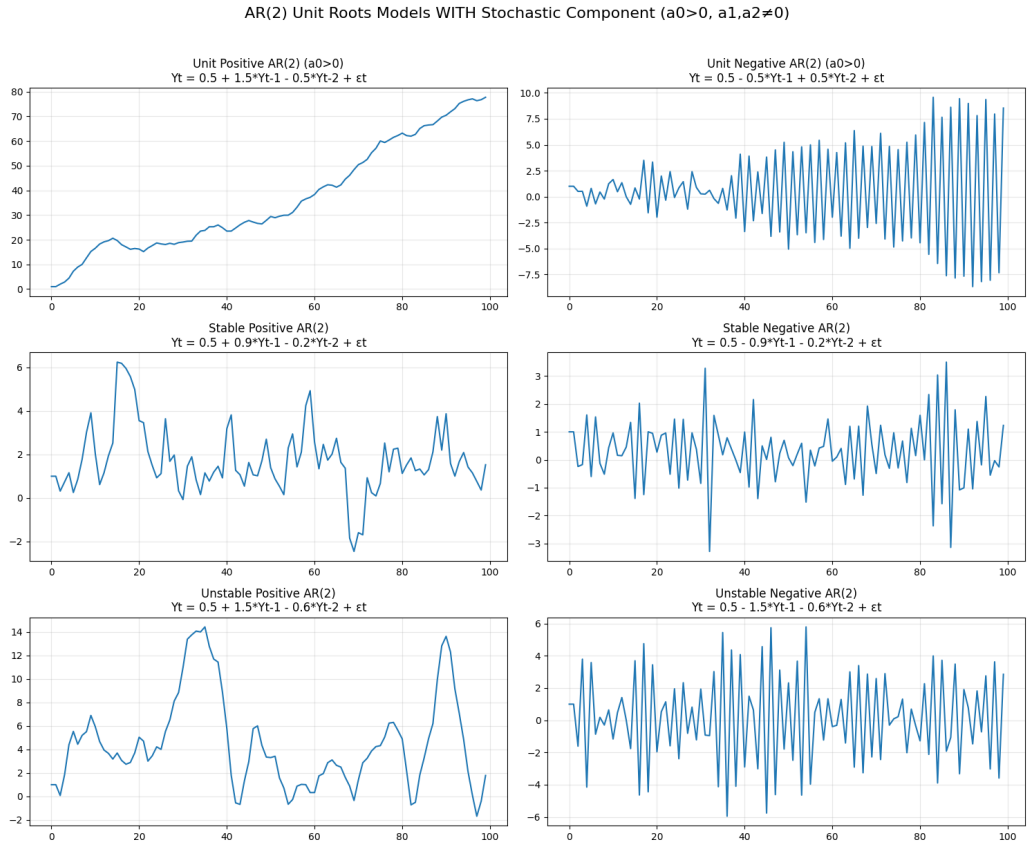

Figure 4: AR(2) process with stochastic component

1.5 Bivariate Time Series

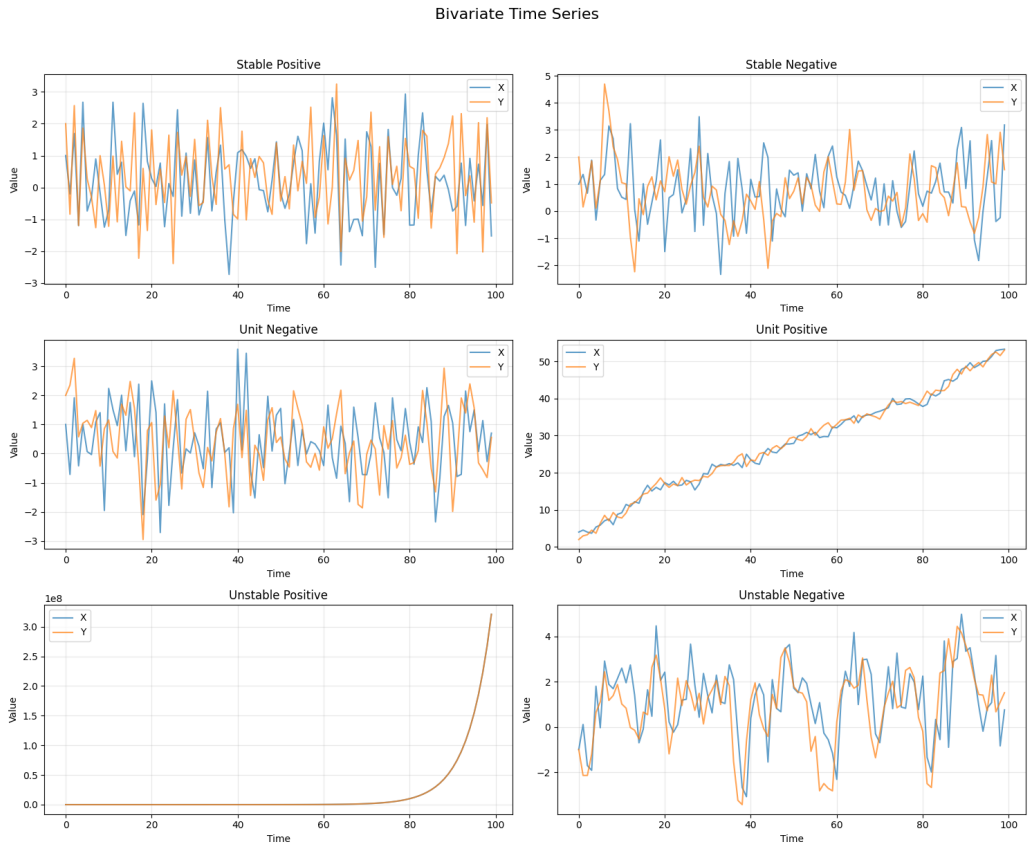

Figure 5: Bivariate time series models

2 Task 2

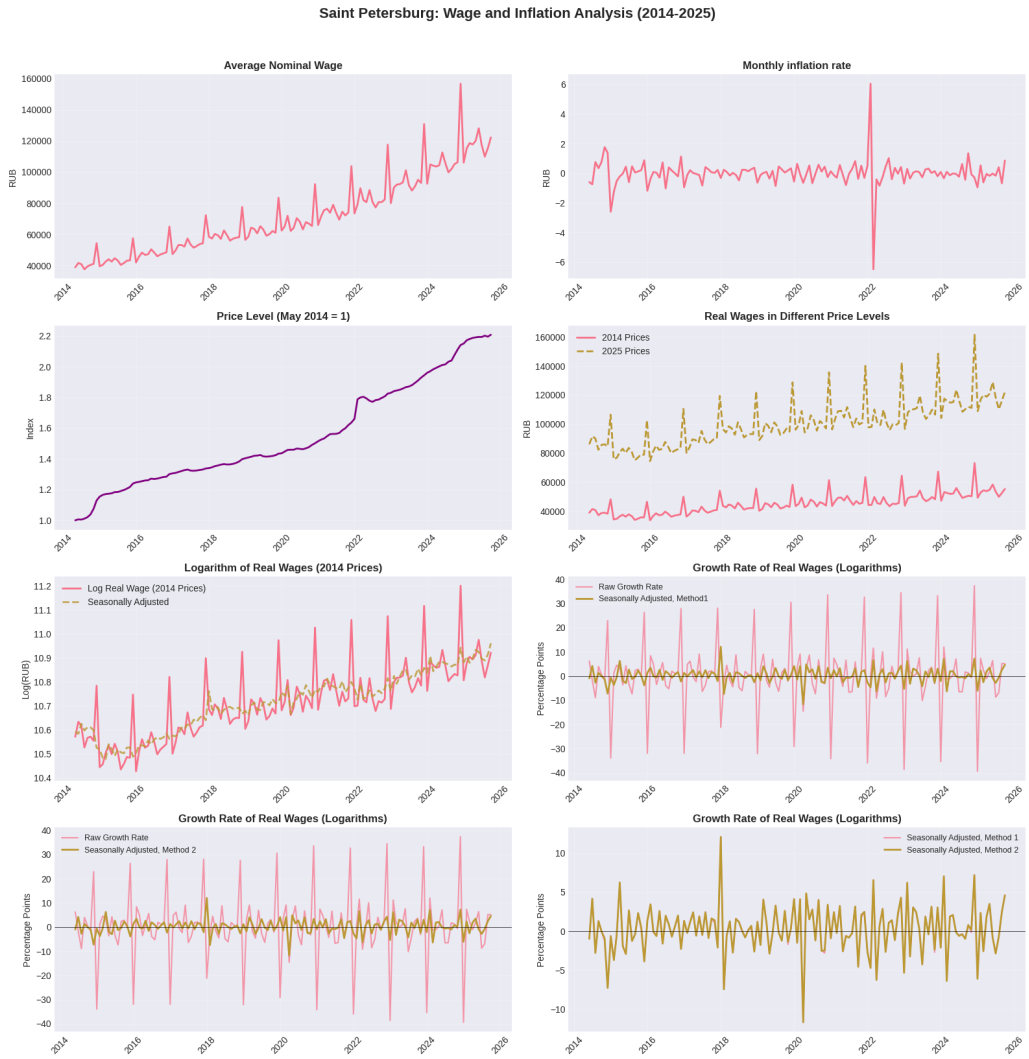

Figure 6: Saint-Petersburg Wage and CPI

3 Task 4

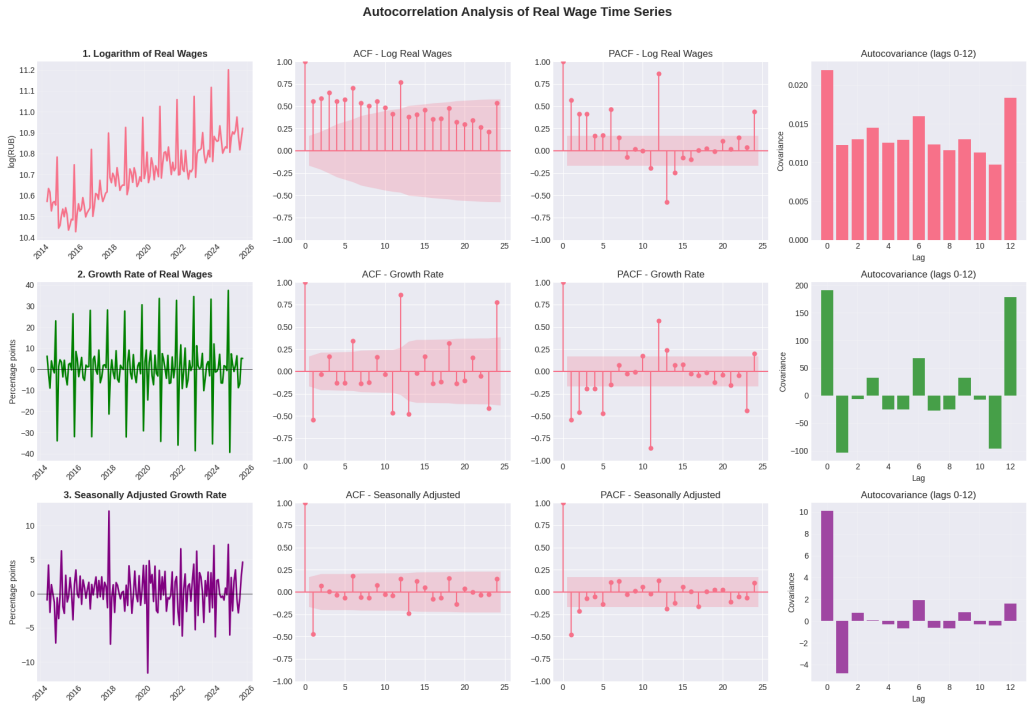

Figure 7: PACF (Partial Autocorrelation Function)

Ljung-Box Test Results

Table 1: Ljung-Box test for autocorrelation

Series	**Lags**					
	1	2	3	6	12	
Log of Real Wages	43.93 (0.000)	93.23 (0.000)	153.68 (0.000)	318.66 (0.000)	596.74 (0.000)	87
Growth Rate	40.98 (0.000)	41.12 (0.000)	45.03 (0.000)	66.75 (0.000)	221.98 (0.000)	42
Seasonally Adjusted	31.42 (0.000)	32.14 (0.000)	32.15 (0.000)	37.66 (0.000)	43.24 (0.000)	6

Note: Values show test statistic (p-value). All tests reject the null hypothesis of no autocorrelation at all lags.

Interpretation:

- **Logarithm of Real Wages:** Significant autocorrelation detected at all tested lags
- **Growth Rate of Real Wages:** Significant autocorrelation detected at all tested lags
- **Seasonally Adjusted Growth Rate:** Significant autocorrelation detected at all tested lags

4 Task 5

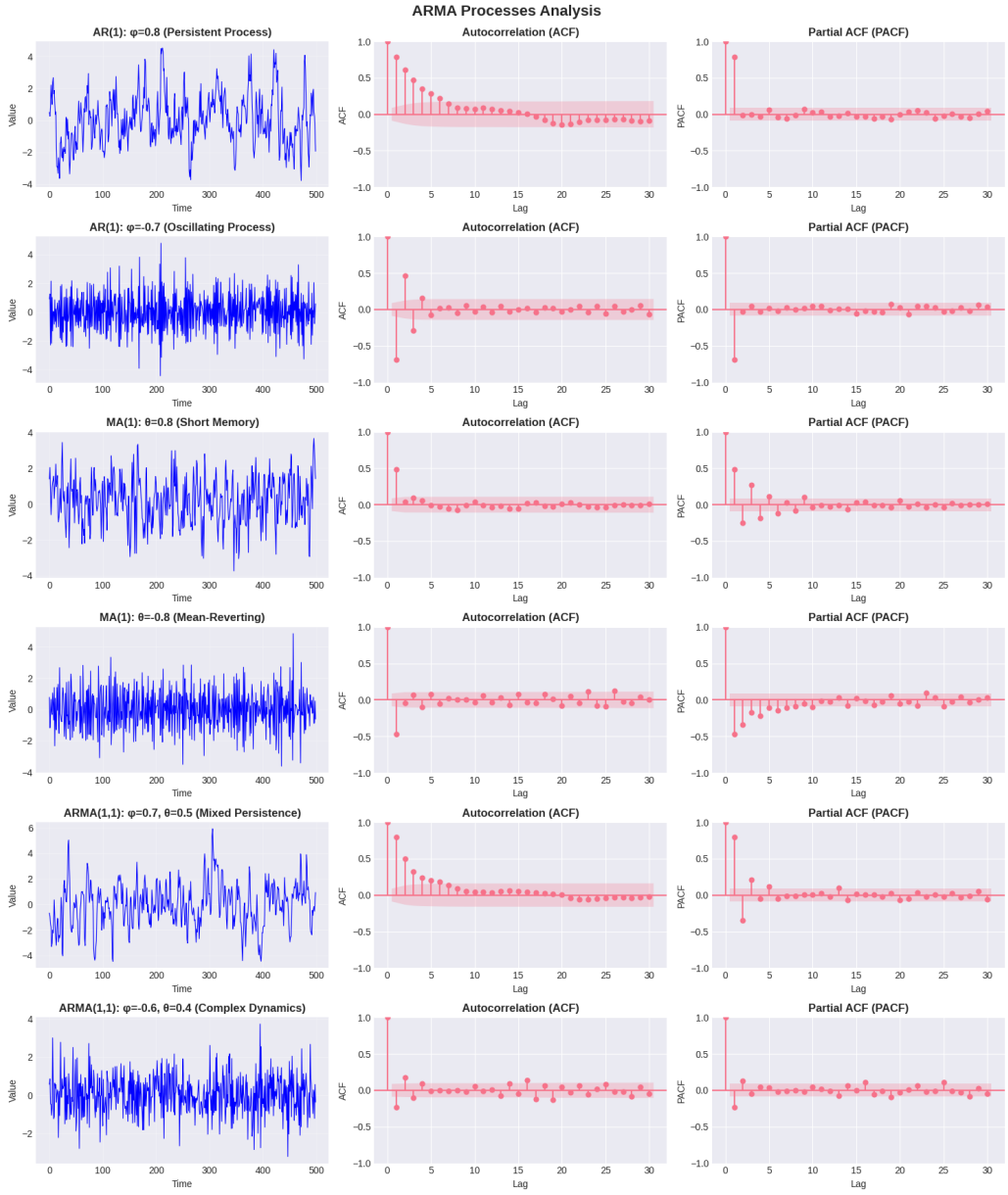

Figure 8: ARMA model results